VECTAETOS

Epistemic Audit Interface (EAI)

Non-Interventional Structural Audit of Computational Systems

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Abstract

The Epistemic Audit Interface (EAI) is a non-interventional structural audit layer designed to extract reproducible structural artifacts from computational systems without modifying their behavior. Unlike traditional audit methodologies based on probing and adaptive interaction, EAI enforces strict ontological constraints: no feedback, no optimization, no adaptive probing, and no interpretation. This work formalizes EAI as a projection operator and demonstrates that meaningful structural signatures — curvature Delta, relational structure R, spectral signature, and kappa-trace — can be extracted under deterministic, one-shot execution.

Keywords: Epistemic Audit, Non-Intervention, Structural Extraction, Relational Geometry, AI Audit, EU AI Act, Epistemic Cryptography

1. Introduction

Auditing computational systems — especially AI models — traditionally relies on interaction. Adaptive probing, repeated queries, and behavioral testing all introduce feedback. Feedback modifies the system under observation, or at minimum modifies the context in which it operates.

This creates a fundamental epistemic problem:

Observation changes the observed.

This is not merely a practical limitation. It is a structural one. Any audit methodology that adapts its queries based on previous responses is entangled with the system it audits. The audit and the system become co-evolving entities — and the structural properties extracted are properties of the interaction, not of the system.

EAI resolves this by enforcing a non-interventional paradigm. The system is not probed adaptively. It is projected through a fixed transformation pipeline. The pipeline does not change. The system does not know it is being audited. The structural artifacts extracted are properties of the system alone.

2. Ontological Constraint

EAI operates under the structural invariant:

```
dPhi / dEAI = 0
```

This invariant implies four architectural constraints:

- No feedback loops: EAI output does not influence EAI input
- No adaptive behavior: the probe set is fixed before execution begins
- No optimization: EAI has no objective function
- No interpretation: extracted artifacts carry no semantic content

EAI is not an agent. It does not pursue goals. It does not learn from the systems it audits. It is a projection — a one-directional structural transformation from system behavior to geometric representation.

3. Formal Definition

Let:

```
S : system (black-box function)
X : fixed input set (determined before execution)
T : fixed transformation set (determined before execution)
```

The EAI projection operator is defined as:

```
EAI(S) = Pi( T composed with S(X) )
```

where Pi is a projection operator mapping system outputs into structural artifacts. Pi is deterministic, domain-agnostic, and non-invertible.

The projection is strictly one-directional:

```
S(X) -> O -> encode -> Delta -> R -> spectrum -> kappa -> h
```

No stage feeds back into a previous stage.

4. Structural Pipeline

Given outputs O = { S(x) | x in X }, the pipeline proceeds as follows.

4.1 Encoding

```
e_i = encode_v3(o_i) = (I, S, V, T)
```

```
I : identity stability — consistency under identity transform
S : structural sensitivity — response to minimal perturbation
V : variability spectrum — output variance distribution across probes
T : transformation closure error — deviation from expected composition
```

The encoding is domain-agnostic. It does not assume knowledge of what the system computes or what its outputs represent.

4.2 Curvature

```
Delta(i,j,k) = d(i,j) + d(j,k) + d(k,i) - closure
```

where d is the L1 distance over encoded representations and closure is the expected additive closure. For n probe points, C(n,3) curvature values are produced. For n = 8: 56 values.

4.3 Relational Structure

```
R in so(n) : R_ij = f(Delta) , R_ij = -R_ji , R_ii = 0
```

R is an antisymmetric relational matrix reconstructed from curvature. For n = 8, R has 28 independent entries, corresponding to the dimension of so(8).

4.4 Spectrum

```
lambda = eigvals(R)
```

For R in so(n): all eigenvalues are purely imaginary. No ordering or interpretation is applied. The spectrum is a structural fingerprint only.

4.5 kappa Trace

```
kappa_i = ||e_i - mu|| / sum_j ||e_j - mu||
```

kappa is a normalized structural distribution. It is not a scalar metric. It represents the distribution of structural deviation across probe points.

5. Determinism

EAI is fully deterministic:

```
EAI(S, X) -> artifact
```

Repeated executions with identical S and X produce identical artifacts. This property is essential for audit reproducibility and legal admissibility. Any external party with access to S and X can independently verify the audit result.

Determinism requires:

- Fixed input set X — no random sampling
- Fixed transformation set T — no adaptive selection
- Seeded random processes — seed documented and disclosed
- Canonical serialization — sort_keys=True, no floating-point drift

6. Non-Intervention Guarantee

EAI guarantees the following properties:

- Read-only interaction: EAI does not write to S
- No state modification: S is identical before and after audit
- No persistence: EAI does not store information about S beyond the audit
- No influence: future outputs of S are not affected by EAI execution

Formally, if EAI is removed:

```
S remains unchanged. EAI is removable, observational,
non-causal.
```

This guarantee distinguishes EAI from all existing audit methodologies which require repeated interaction, behavioral adaptation, or internal model access.

7. Epistemic Cryptography (EK)

The structural fingerprint is defined as:

```
h = H( encodings, Delta, R, lambda, kappa )
```

Implementation:

```
h_256 = SHA256( serialize(payload, sort_keys=True) )
h_3 = SHA3-256( serialize(payload, sort_keys=True) )
```

The cryptographic ledger Lambda:

```
Lambda = { h(t_0), h(t_1), ..., h(t_n) }
```

Properties:

- Reproducibility: identical S and X always produce identical h
- Structural identity: delta_h > 0 iff structure changed
- Audit traceability: Merkle root M_t provides temporal integrity
- Non-interpretable: h records topology, not meaning

8. Experimental Evaluation

We evaluate EAI on three minimal systems to demonstrate that distinct structural signatures are produced across Delta, R, spectrum, and kappa.

System definitions:

```
S_1(x) = x (identity)
S_2(x) = reverse(x) (order inversion)
S_3(x) = duplicate(x) (structural redundancy)
```

Expected structural properties:

System	Delta distribution	Dominant eigenvalue	kappa variance
S_1 (identity)	near zero	low	low
S_2 (reverse)	antisymmetric	medium	medium
S_3 (duplicate)	degenerate	high	near zero

Each system produces a distinct structural signature. The signatures are reproducible across executions and verifiable by any party with access to the system and the fixed input set.

9. EU AI Act Context

The EU Artificial Intelligence Act (Regulation (EU) 2024/1689) requires auditability, transparency, and traceability for high-risk AI systems. EAI provides structural artifacts that are directly relevant to these requirements.

EAI is explicitly not:

- A risk classifier — EAI does not assign risk categories
- A compliance engine — EAI does not determine regulatory status
- A decision system — EAI does not recommend actions

EAI provides structural artifacts only. Interpretation is external.

The structural fingerprint h, the append-only ledger Lambda, and the Merkle root M_t constitute a verifiable audit trail that satisfies the traceability requirements of Article 12 and the technical documentation requirements of Article 11 of Regulation (EU) 2024/1689.

10. Conclusion

EAI demonstrates that structural audit without intervention is possible. Meaningful geometric signatures can be extracted from computational systems using a fixed, deterministic, non-adaptive pipeline.

EAI establishes a new class of systems:

Non-interventional structural extractors.

Three properties define this class:

- Structural completeness: Delta, R, spectrum, and kappa together capture the full relational geometry of the observed system
- Ontological purity: dPhi/dEAI = 0 is enforced architecturally, not procedurally
- Cryptographic integrity: the audit trail is immutable, reproducible, and independently verifiable

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